

4. Decompose x into a linear combination of vectors (with numeric entries) using the free variables as parameters.

Practice Problems

1. Each of the following equations determines a plane in $\mathbb{R}^3$. Do the two planes intersect? If so, describe their intersection.

$$\begin{aligned} x_1 + 4x_2 - 5x_3 &= 0 \\ 2x_1 - x_2 + 8x_3 &= 9 \end{aligned}$$

Solution: Two planes intersect if and only if the given system of equations has a solution set.

$$x_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} 4 \\ -1 \end{bmatrix} + x_3 \begin{bmatrix} -5 \\ 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 9 \end{bmatrix}$$

By defn. of $Ax=b$, we can rewrite this equation as

$$\begin{bmatrix} 1 & 4 & -5 \\ 2 & -1 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 9 \end{bmatrix}$$

By Theorem 3, the above equation $Ax=b$ has the same solution set as the system of linear equations corresponding to the augmented matrix $[A \ b]$

$$\begin{bmatrix} 1 & 4 & -5 & 0 \\ 2 & -1 & 8 & 9 \end{bmatrix} \xrightarrow[\text{Row 2 = Row 2}]{\text{Row 2 = Row 2} - 2 \times \text{Row 1}} \begin{bmatrix} 1 & 4 & -5 & 0 \\ 0 & -9 & 18 & 9 \end{bmatrix} \xrightarrow[\times (-1/9)]{\text{Row 2 = Row 2}} \begin{bmatrix} 1 & 4 & -5 & 0 \\ 0 & 1 & -2 & -1 \end{bmatrix}$$

Verification of whether the final matrix is in rref form

- Row 1 and 2 are both non-zero rows. $\therefore$ The condition that all non-zero rows must be above the zero rows is vacuously True.
- Row 1 Leading Element = 1 in Column 1
Row 2 Leading Element = 1 in Column 2
 $\therefore$ The condition that for each row, the leading element must be in a column to the right of the columns of the leading elements of all rows above it is True.